`$ mysql -e "SHOW CREATE TABLE orders" --normalize`

□□□□□□

// 1NF → 2NF → 3NF □□□□□

1NF □□□	**2NF** □□□□□□	**3NF** □□□□□□	**□□□□** OLAP □□□

$ cat schema-plan.md

01 /

02 /

1NF → 2NF → 3NF

03 /

JOIN

04 /

OLTP vs OLAP

——

PART 01

[illegible]

order_id	customer_name	customer_email	product_name	price
1	[illegible]	zhang@example.com	[illegible]	30000
2	[illegible]	zhang@example.com	[illegible]	500
3	[illegible]	li@example.com	[illegible]	1200

[illegible]

[illegible] email [illegible]

[illegible]

[illegible] email[illegible] UPDATE [illegible]

→

ROOT CAUSE

CHECKPOINT_01

[illegible] UPDATE 37 [illegible] 3 [illegible]

- A [illegible]
- B [illegible]
- C [illegible]

PART 02

[illegible]

1NF → 2NF → 3NF[illegible]

1NF [illegible]

BEFORE — [illegible]

student	courses
[illegible]	[illegible], [illegible], [illegible]

AFTER — 1NF

student	course
[illegible]	[illegible]
[illegible]	[illegible]
[illegible]	[illegible]

[illegible]

[illegible]

[illegible]

2NF□□□□□□□

□□□ (order_id, product_id)□□ product_name □□□ product_id —— □□□□□□□□

BEFORE — □□□□

order_id	product_id	product_name	qty
1	P1	□□	1
2	P1	□□	3

AFTER — □□□□□

order_id	product_id	qty
1	P1	1

product_id	product_name
P1	□□

□□□□□□□□□**□□□□**□□□□□□□□□□□□□

3NF□□□□□□□

student_id → dept_id → dept_name□dept_name □□ dept_id □□□□□□ —— □□□□□□□

BEFORE — □□□□□

student_id	name	dept_id	dept_name
S1	□□	D1	□□□
S2	□□	D1	□□□

AFTER — □□□□□□

student_id	name	dept_id
S1	□□	D1

dept_id	dept_name
D1	□□□

□□□□□□□□□□□□□ —— □□□□□□□□□□□□□

1NF

2NF

3NF

3NF

PART 03

[illegible]

[illegible]JOIN [illegible]

JOIN

JOIN

OLTP vs OLAP□□□□

□□□□	□□□	□□□□
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□□□□	**OLTP** □□□□	**OLAP** □□□□□□□

STRATEGY

□□□□□□□□□□□□□□□□□□□□□□□□□□□ — □□□□□□□□□□□□□□□□□□□□□□□□□□□□

→

1NF → 2NF → 3NF

3NF

OLTP OLAP

Schema normalized.

[illegible] JOIN [illegible]

```
$ mysql -e "EXPLAIN SELECT ..."
1NF → 2NF → 3NF → [illegible]
```